Review of the algorithms steganography in PDF documents and analysis

Alexander V. Sergeev, Pavel. B. Khorev, *National Research University "MPEI"*

***Abstract*—In this article, we will explore methods of using data hiding in a pdf document. The widespread use of PDF files can make its use for this purpose an interesting and practical solution. We was explore of algorithms of steganography algorithms from open sources and based on the analysis, an algorithm for detecting steganography proposed. We explore the methods to use a hide data in structure of the pdf document format, the basic strategies for hiding data, the statistical characteristics of PDF operators, we analysis the detection of changes after applying the algorithm for hiding data.**

***Keywords*—Data transfer, Steganography, Information security**

I. Introduction

Steganography is a science that studies the ways and methods of hiding sensitive information. One of the occurrences of information leakage is the use of steganographic tools, that is, methods for transferring or storing information that hide not only the contents of a secret message but also the fact of its existence. Typically, a message may look like something else, such as an image, an article, a shopping list, or a letter.

In connection with the development of computer hardware and a huge number of information transfer channels, new steganographic methods based on the features presenting information in files, computer networks, and other locations with the use of special software or standard tools, have appeared.

Most of the steganographic containers are images, video clips, music, sounds [7], and texts. But text steganography is the most complicated due to the lack of redundant information in text files, while images or sound files contain a lot of natural redundancy, which leads to the high efficiency of using these files in steganography.

II. The structure of the PDF

A. Selecting a Template (Heading 2)

The PDF format created by Adobe Systems [1] for document exchange is a fixed layout format for presenting documents in a way that is independent of application software, hardware, or operating system. Currently, PDF files are often used and this fact allows them to be applied as accompanying documents when hiding information. There are very few studies offering methods for using these files as information-hiding objects (steganographic containers).

III. Methods of hiding data

When analyzing existing methods of applying steganography in PDF files, the White Space Coding algorithm, which uses regular space and non-breakable space visually indistinguishable but with different codes, may be isolated [2]. In the TJ method (Text Justified, a text-aligning operator method) [3], hidden data is embedded in the gaps between words and the text-positioning operator is used [5, 6]. When aligning, the lengths of the gaps take a random character, which makes it possible to modify them using LSB (Least Significant Bit) methods.

In another algorithm based on the paper by I-Shi Lee [2], secret data is embedded between words or between character locations in a PDF file using non-breakable space with A0 code of the American standard information exchange code (ASCII). The authors, in their study, found out that the non-breakable space (A0), being embedded in a string of text characters, is a symbol. It becomes invisible in several versions of ordinary PDF readers and this phenomenon may be used to hide data.

This paper notes that after the message is embedded, two files are transferred (a file containing Huffman codes for secret message characters, and a PDF file with a secret message).

In the paper [4], using this principle, new algorithms have been proposed when working with gap encoding. It is suggested to use a gap (code 0x20) and a non-displayable space (code 0xA0) as a binary sequence of a secret message (b) according to the following principle:

- If b=1; then replace 0x20 between words with 0xA0
- If b=0; without changes

The use of the algorithm based on a non-breakable space may be used only in specific cases and its steganographic properties do not have the proper durability since this method only works with Adobe Reader of version 6.0 Pro, 7.0 Pro, 8.1.2, and 9.0 in the environment of Windows OS. If a document is opened using another PDF reader, the changes in the PDF document will be visible.

The algorithm based on the papers by Jiun-Tsung Wang [5], Yin-Cheng Lai [6] uses the features of displaying text blocks on the pages of a PDF document. According to the PDF standard, PDF files contain eight main types of objects. One of them is the page object, which includes options for describing the properties of the page in a PDF file. The authors focus on the text matrix operator.

$$T_m = T_{LM} = \begin{bmatrix} a & b & 0 \\ c & d & 0 \\ e & f & 1 \end{bmatrix} (1)$$

$$T_m = \begin{bmatrix} a & b & c & d & e & f \end{bmatrix} (2)$$

The text matrix consists of six different numbers, a, b, c, d, e, and f, as well as of the Tm operator. The structure of the text matrix is shown in Formulation 1. The structure of the linear matrix is shown in Formulation 2. The values of a, b, c, and d are the elements of the matrix that are used to describe the scale and orientation of the text object, and the numbers e and f are the conversion parameters that are used to descriptions of the text position. The authors of the paper

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[5, 6] propose a method for embedding sensitive data in the parameters e and f of numerical objects in a PDF file.

The algorithm [6] assumes that the position of the text in the text matrix is outside the visible area of the displayed PDF document. To do this, "999" is added in front of the values of e and f of each text matrix created. That is, the hidden secret message is encoded and replaces the values of e and f. The final text matrix would take the following form: [1 0 0 1 999327 0] Tm

In the algorithm proposed in [5], it is supposed to use changes in the text matrix after the sign of the values of e and f. In this case, 2 methods are suggested: 1) addition of N zeros, this method will not change the document data but will affect its size. For example, the number 20 will change to 20.00000. In method 2, the message is encoded into a decimal code and added after the decimal point. For example, the number 20 will change to 20.00457.

If the values of e or f already have a decimal part, two zeros “00” then are added. For example, 854.7 will change to 854.700457.

The visible area of the MediaBox page is described by four numbers, with the first two numbers indicating the position of the upper left corner and the last two numbers indicating the position of the lower right corner. These four numbers determine the width and height of the visible area of the page. The page object that includes MediaBox is shown in Figure 1.

```
11 0 obj
<<</Type/Page
/Parent 1 0 R
/Resources 14 0 R
/MediaBox[0 0 595.2 841.68]
/Contents[12 0 R ]
>>
Endobj
```

Figure 1. Example of page object

Small changes in the width and height of the page are not so easy to notice, and therefore the authors of the paper [5] propose to insert secret data into the MediaBox of pages). The message is encoded by converting the source text to binary code, it is then divided into 3 bits and the corresponding decimal values are assigned to them. Data is added to the last two parameters . Whereupon, if the number is an integer, then a sequence of encoded messages ".00" followed by a coded message, is added to it. Otherwise, "00" (two zeros) is added and then, a coded message.

The algorithm suggested by Shangping Zhong [3] has described an original algorithm that exploits the "TJ" operator (taking into account the clarity of text perception): a set of numbers are selected from the strings of the "TJ" operator and hidden data is embedded in them.

IV. ANALYSIS OF CHANGES

PDF contains various operators that may be used to display text, as well as to adjust the position of the text within a PDF document.

Tc operator. This operator is used to control the gap between characters and extends to the entire text block. The Tw operator is used to set a gap between words.

It works identically to the Tc operator but attributed only to the gap character.

The TJ operator is used to display text strings in a PDF file. It contains an array of strings and numbers, which consist of characters and gap values respectively, which are used between these characters. For each TJ value, the current text position changes by subtracting the value from the current position. The results of the analysis of 8 PDF documents [8-16] for the number of operators used are shown in Table 1.

Table 1

Statistical Characteristics of PDF Operators

	1	2	3	4	5	6	7	8
TJ	368	368	1085	1144	1205	3098	3461	803
Tm	391	437	467	488	533	2224	2579	5874
Tc	1632	11687	12746	13527	13671	16599	17005	25037

The TJ operator, compared to the Tc and Tw operators, is used in every PDF file. Each string of the text is represented by a single TJ operator. Each TJ statement contains one or more document space values. If the text is aligned in width, which means that it is aligned with the left and right margins, the TJ operator is used more often to input a variable distance between words and characters to comply with alignment rules. On the contrary, Tc and Tw values contain only one document space value for a block of text. Although the Tc and Tw operators may be used to hide data in PDF documents, TJ values seem to be the most promising.

Based on the principle of describing a page by the TJ operator, it has been confirmed that the PDF text will not change its perception if we change only a small part of the TJ operator integers, which are responsible for the position of characters in the PDF text. A secret message is embedded using the LSB (the least bit) technology into the values of Tj [-16;16].

One of the important parts of the original algorithm is the use of logistic mapping as a source of random numbers. In one case, it is used to select a random location for embedding data, and in another, it is used to create random numbers between [-16;16], which may be inserted to create redundancy.

The studies conducted (results in Figures 2, 3, 4 and 5) show that the changed values in the PDF document created using this algorithm may be detected by considering the distribution of TJ values in the PDF file.

As a result of the analysis, it may be concluded that the use of additional methods does not complicate the detection of the existence of hidden data when using statistical analysis.

From the results of the frequency analysis, it follows that the hidden data is likely to be detected because the TJ values are replaced by random values generated by a pseudo-random number generator, which distributes them to equivalent values of the frequencies of this range.

To assess the size ratio of the hidden secret message and the PDF document, an image of 4,668 bytes has been embedded in a PDF document of 24,026 bytes, which, therefore, indicates the possibility of embedding messages of

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about 19.43% of the size of the PDF document (steganographic container).

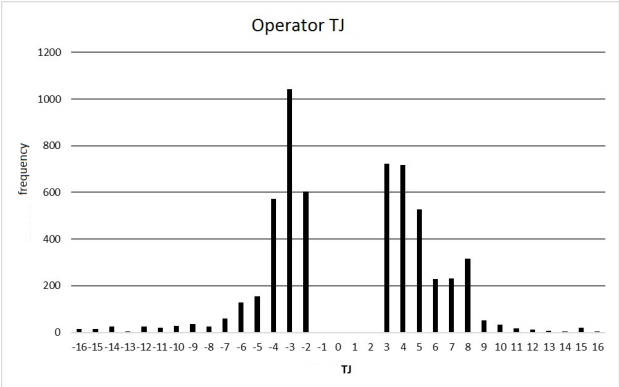

Figure 2. The resulting operator values on the interval [-16; 16]

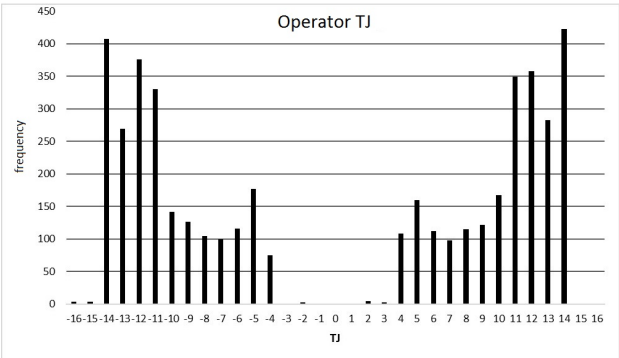

Figure 3. The document after applying the algorithm

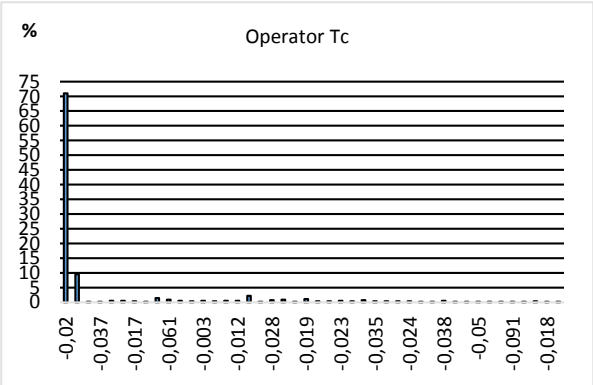

Figure 4. The obtained values of the operator Tc

Fig. 4 shows the values of the Tc operator. As you can see, ~ 80% are occupied by values of -0.02 and -0.025. The use of integer or fractional values of the Tc operator, rounded to the nearest hundredth, will be determined in the PDF document. Accordingly, if we use this operator for further processing, then the number of Tc operator parameters values will significantly decrease (for the document under consideration, to 109).

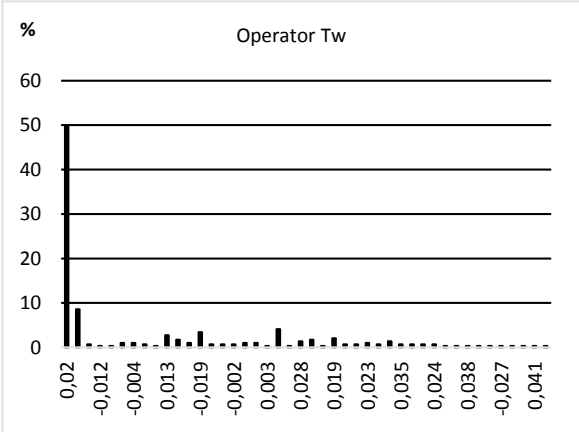

Figure 5. The obtained values of the operator Tw

Fig. 5 shows the parameters of the Tw operator. As you can see, ~ 50% are occupied by values of 0.02. The figure shows the text with the original data, the first line [0.0651 Tw] and replaced by [0.02 Tw] second line. No visible differences were found, which corresponds to one of the properties of steganography.

V. Conclusion

The algorithms reviewed in this paper allow concluding that this field is of significant interest and is gaining popularity so far. Each of the reviewed algorithms has its advantages and disadvantages:

According, an analysis of the results obtained makes it possible to detect data changes when hiding a message in a PDF document by mathematical analysis of the data.

As can be seen from the Table 1, the Tj operator has a predominant number of parameter values, but due to the need to control entropy in the PDF document used as a steganographic container, the maximum volume of hidden message is reduced.

We propose to consider the method of mixed message embedding, which allows apply all the operators used in a PDF document.

The combined approach is proposed to be used as follows:

1) Tj has the most hidden attachment of secret data. It is possible to detect data changes by a mathematical analysis of the data.

2) A0 can work under harsh conditions, which, if not observed, will make a steganographic container detected.

3) Tm, using "00" to hide a message will affect the file size, and during analysis, a change in standard values will be detected. The algorithm with the addition of coded values will not affect the change in the appearance of the document. No mathematical analysis has been performed for this algorithm and its durability may not be assessed.

4) MediaBox, the number of objects where this operator is present, is much less than Tm. It is reasonable to be used when hiding short messages or in documents with a large number of sheets.

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